

PHASE 6 – PROJECT DEVELOPMENT PHASE

Objectives

- To develop and implement the Medical Inventory Management System on the Salesforce platform.
 - To build custom objects, automation, and reports as per the project design.
 - To ensure all functionalities work efficiently and meet user requirements.
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Development Activities

1. Custom Object Creation

- Created custom objects: *Medicine*, *Supplier*, *Purchase Order*, and *Inventory Transaction*.
- Added necessary fields such as *Item Name*, *Batch Number*, *Expiry Date*, *Quantity*, and *Reorder Level*.
- Established relationships between objects using lookup fields.

2. Page Layouts and Record Types

- Customized page layouts for each object to display relevant information.
- Created record types where needed (for example, *Medicine* and *Equipment* types).

3. Automation and Workflows

- **Flows:**
 - Stock quantity automatically updates after a new transaction.
 - Expiry alert flow triggers emails for items nearing expiry.
- **Approval Process:** Implemented for purchase orders requiring manager approval.
- **Validation Rules:** Prevent negative quantities and expired item entries.

4. Apex Development

- Used Apex Triggers for updating stock counts.
- Created Test Classes to validate automation logic and maintain code coverage.

5. Reports and Dashboards

- Built custom reports for:
 - Current Stock Levels
 - Expired Items
 - Low-Stock Medicines

- Created dashboards for visual tracking of inventory metrics and supplier performance.

6. User Interface Design

- Customized Lightning App with tabs for *Medicines*, *Suppliers*, *Orders*, and *Reports*.
- Added quick actions for “Add Stock” and “Issue Stock”.
- Used Lightning Pages to show related data on a single screen.

7. Security Setup

- Configured Profiles and Permission Sets for Admin, Pharmacist, and Inventory Manager.
- Applied field-level security and record-sharing settings.

8. Data Import

- Imported sample data for medicines and suppliers using Salesforce Data Loader.
- Verified data integrity and accuracy after import.

Testing During Development

- Unit testing of Flows, Validation Rules, and Triggers.
- Verified record creation, updates, and automation behavior.
- Ensured that alerts and reports function correctly.
- Fixed bugs found during testing.

Tools and Resources Used

- **Developer Console / VS Code** – for coding and deployment.
- **Data Loader** – for importing and exporting data.
- **Reports and Dashboards** – for analytics and visualization.
- **Flow Builder** – for automation and process management.

6.5 Expected Outcomes

- Fully functional Salesforce-based inventory system.
- Automated stock tracking and expiry alert processes.

- Accurate reporting and dashboard insights.
- Role-based secure access for all users.